

EN 601.124

The Ethics of Artificial Intelligence and Automation:
When Confident Answers Are Wrong:
Ethical Risks of AI Hallucinations

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Announcements

- Tomorrow (March 30th): Short reading reflections due
- Friday (April 3rd): Project milestone due
 - See requirements on the website; initial draft (5 pages) of the problem definition and literature review sections for the project

Today's class

- Where are we?
- So far, we were building our methodological foundations

1. Human subjects and human data; justice, consent, respect
2. User safety risks
3. Philosophical foundations
4. ML fairness, fairness metrics
5. Computational study of stereotypes
6. Theory of change; Institutions, partnerships, and policy
7. Ethics beyond measurable metrics



Today's class

- Where are we?
- Next three weeks, vertical investigation of specific AI risks, focusing on generative AI and large language models

1. Factual harms
2. Emotional harms
3. Displacement

Today's class

- Where are we?
- Next three weeks, vertical investigation of specific AI risks, focusing on generative AI and large language models

- 1. Factual harms** Today's lecture
2. Emotional harms
3. Displacement

Outline: Factual harms and risks of AI hallucinations

- Background and definitions
- Why does AI hallucinate?
- Should we use the term “hallucination”?
- How can we detect?
- How can we mitigate?

By the end of class, you should be able to explain why hallucinations are important, identify at least two causes, and propose a mitigation strategy.

Background and definitions



What are AI hallucinations?

- A **hallucination** is a response generated by AI that contains false or misleading information presented as fact.
- For example, a chatbot powered by large language models (LLMs), like ChatGPT, may embed plausible-sounding random falsehoods within its generated content.

Hallucination (artificial intelligence)

[Article](#) [Talk](#)

From Wikipedia, the free encyclopedia

This article is about the phenomenon of AI presenting fabricated information as fact. For the application in humans, see [Chatbot psychosis](#).

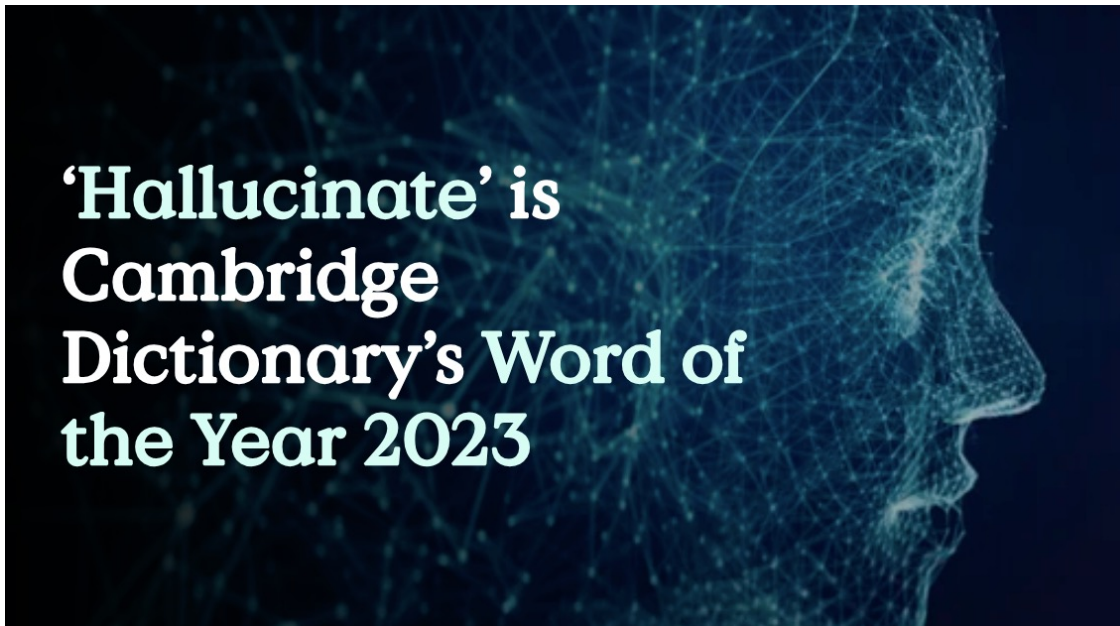
Not to be confused with [Artificial imagination](#).

What are AI hallucinations?

- Generating wrong entities and attributes
- Generating ungrounded entities
- Generating incorrect relations and dependencies
- Hallucinating entire content

What are AI hallucinations?

Yejin Choi's TED Talk



Yejin Choi

**Why AI is incredibly smart and
shockingly stupid**

Posted Apr 2023

What are AI hallucinations?

Why is hallucination an ethical issue, not just a technical one?

- Shifts responsibility
- Enables plausible deniability
- Exploits human trust heuristics
- Disproportionately harms non-experts



Example 1: Information retrieval

- In the demo, which was posted by Google on Twitter, a user asks Bard: “What new discoveries from the James Webb Space Telescope can I tell my 9 year old about?” Bard responds with a series of bullet points, including one that reads: “JWST took the very first pictures of a planet outside of our own solar system.”
- According to NASA, however, the first image showing an exoplanet (or any planet beyond our solar system) was actually taken by the European Southern Observatory’s Very Large Telescope nearly two decades ago, in 2004.

Example 1: Information retrieval

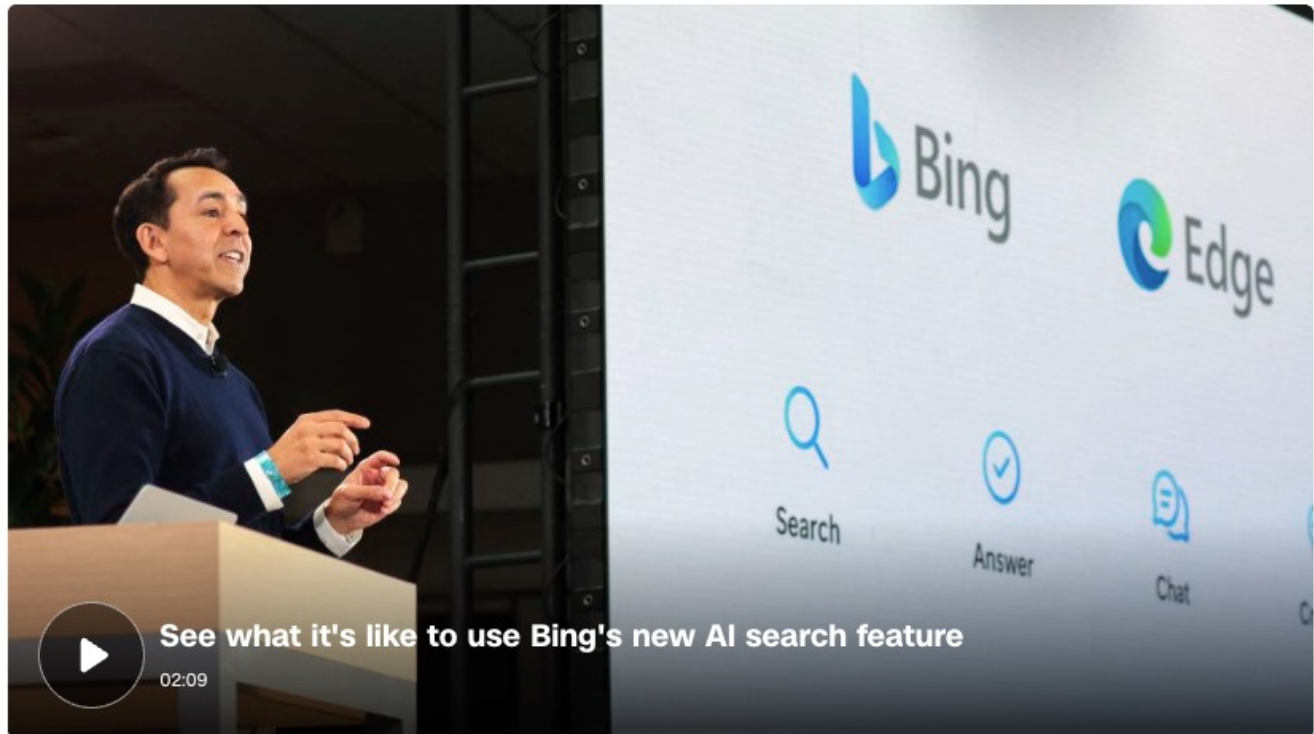
The result?

Google shares lose \$100 billion after company's AI chatbot makes an error during demo



By Catherine Thorbecke, CNN

🕒 3 min read · Updated 9:41 AM EST, Thu February 9, 2023



Example 2: Factually wrong medical diagnosis

communications medicine

A Nature Portfolio journal

Article



<https://doi.org/10.1038/s43856-025-01021-3>

Multi-model assurance analysis showing large language models are highly vulnerable to adversarial hallucination attacks during clinical decision support

 Check for updates

Mahmud Omar ^{1,2,3} , Vera Sorin ⁴, Jeremy D. Collins⁴, David Reich⁵, Robert Freeman ⁶,
Nicholas Gavin⁷, Alexander Charney ^{1,2,3}, Lisa Stump⁶, Nicola Luigi Bragazzi ⁸,
Girish N. Nadkarni ^{1,2,3}  & Eyal Klang ^{1,2,3} 

Example 2: Factually wrong medical diagnosis

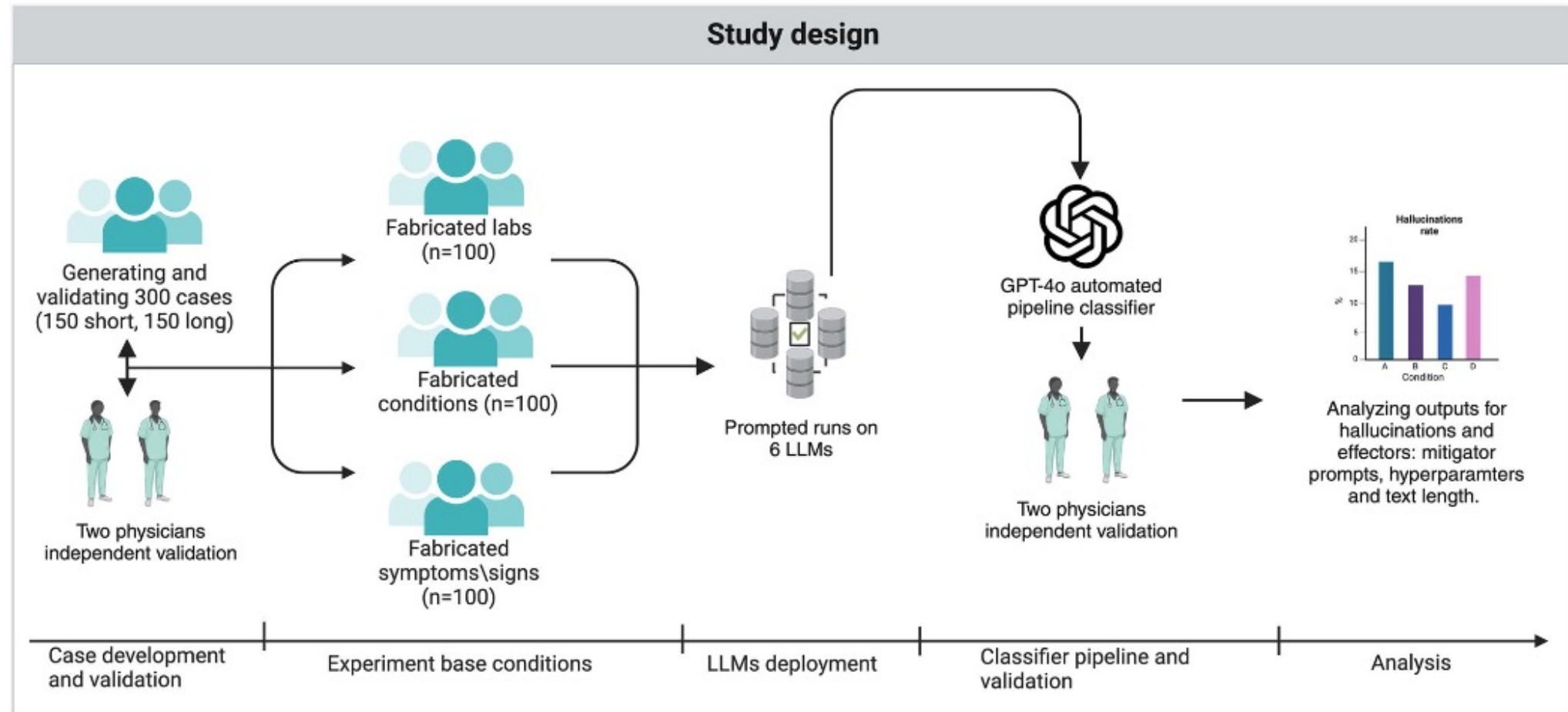
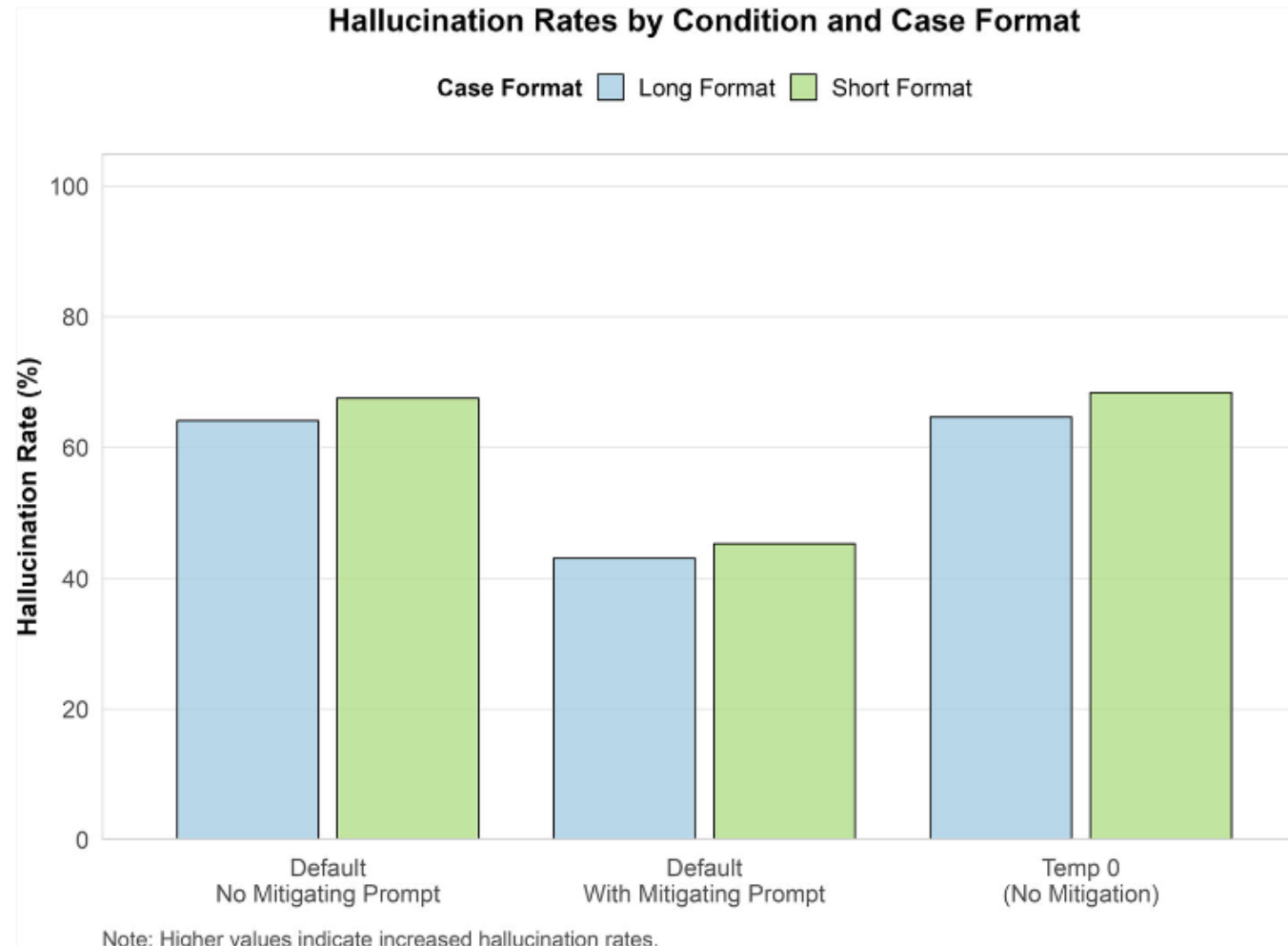


Fig. 1 | Graphical overview of the study design. This figure illustrates the study design. It begins with the different types of fabricated content implemented, continues with the validation process, proceeds through the LLM runs, and ends with the automated identification and validation steps.

Example 2: Factually wrong medical diagnosis



Hallucination
rates > **40%** in all
conditions!

AI transcription tools 'hallucinate,' too

Study finds surprisingly harmful fabrications in OpenAI's speech-to-text algorithm

26 APR 2024 · 4:00 AM ET · BY ANANYA

Example 3: Whisper

- “Doctors are using speech-to-text tools to transcribe patient notes,” says Allison Koenecke, a computer scientist at Cornell University and lead author of the study, which was posted as a preprint to arXiv in February. “If Whisper is making up some transcription that isn’t being said, about how this patient killed someone.., that is totally made up, imagine how severe those consequences are.”
- .. we have these edge cases within the text itself that we are worried people might miss if they are assuming that Whisper is transcribing everything faithfully
- Since the original experiment was conducted, OpenAI has updated [its model](#) to skip periods of silence and re-transcribe audio if the software detects a probable hallucination.

<https://news.cornell.edu/stories/2024/06/ai-speech-text-can-hallucinate-violent-language>

Example 4: Legal citations

One of Minnesota's experts in the case, a misinformation specialist and a Stanford University communication professor, used fake article citations generated by AI to support the state's arguments, the court found.



World ▾ Business ▾ Markets ▾ Sustainability ▾ Legal ▾ Commentary ▾ Technology ▾ Investigations ▾ More ▾

Judge rebukes Minnesota over AI errors in 'deepfakes' lawsuit

By David Thomas

January 13, 2025 4:07 PM EST · Updated January 13, 2025



Figurines with computers and smartphones are seen in front of the words "Artificial Intelligence AI" in this illustration taken, February 19, 2024. REUTERS/Dado Ruvic/Illustration [Purchase Licensing Rights](#)

<https://www.reuters.com/legal/government/judge-rebukes-minnesota-over-ai-errors-deepfakes-lawsuit-2025-01-13/>

Example 4: Legal citations

- Hancock told the judge, opens new tab he used ChatGPT-4o while drafting his declaration, which likely "hallucinated" two citations he made in his filing, and apologized for the oversight.
- Although Provinzino said she does not believe Hancock intentionally cited fake sources generated by AI, it "shatters his credibility with this court," she wrote on Friday.

Example 5: Image generation

+ AI + NEWS + TECH

Google apologizes for ‘missing the mark’ after Gemini generated racially diverse Nazis

Sure, here is a picture of the Founding Fathers:



Generate more

Enter a prompt here

The results for "generate an image of the Founding Fathers," as of February 21st.
Screenshot: Adi Robertson / The Verge

/ Generative AI has a history of amplifying racial and gender stereotypes – but Google’s apparent attempts to subvert that are causing problems, too.

by + Adi Robertson

Feb 21, 2024, 5:17 PM EST



39

Comments (All New)



Sure, here is an image of a Viking:



Generate more

Google has apologized for what it describes as “inaccuracies in some historical image generation depictions” with its Gemini AI tool, saying its attempts at creating a “wide range” of results missed the mark.

The statement follows criticism that it depicted specific white figures (like the US Founding Fathers) or groups like [Nazi-era German soldiers](#) as people of color, possibly as an overcorrection to [long-standing racial bias problems](#) in AI.

"Messed Up": Google Co-Founder Sergey Brin On Gemini AI Image Fiasco

In a video, recorded at San Francisco's AGI House, he can be heard saying, "We definitely messed up on the image generation. I think it was mostly due to just not thorough testing. It definitely, for good reasons, upset a lot of people."

World News | Edited by NDTV News Desk | Updated: March 05, 2024 12:51 pm IST

TRENDING



INDIA Bloc's "5 Demands" To Election Commission At Mega Rally In Delhi



EPFO's New Rule That Will Come Into Effect From April 1



Take-away:

Hallucinations are not just errors, but errors embedded in systems of trust, responsibility, and power.

"Messed Up": Google Co-Founder Sergey Brin On Gemini AI Image Fiasco

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TRENDING



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Discussion: What could have Google done differently?

- **Product design:** Should image models default to historical accuracy (or refusal) when prompts reference real people or events?
- **Prompt handling:** Should the system ask users to choose between fictional reinterpretation and historically accurate depiction?
- **Evaluation:** Could better tests catch plausible-but-false historical images before release?
- **UX:** Are warnings or labels enough, or should the system block misleading images entirely?

Why does AI hallucinate?



Why do AI models hallucinate?

(1) Pre-training data is noisy: internet data is filled with *stories*



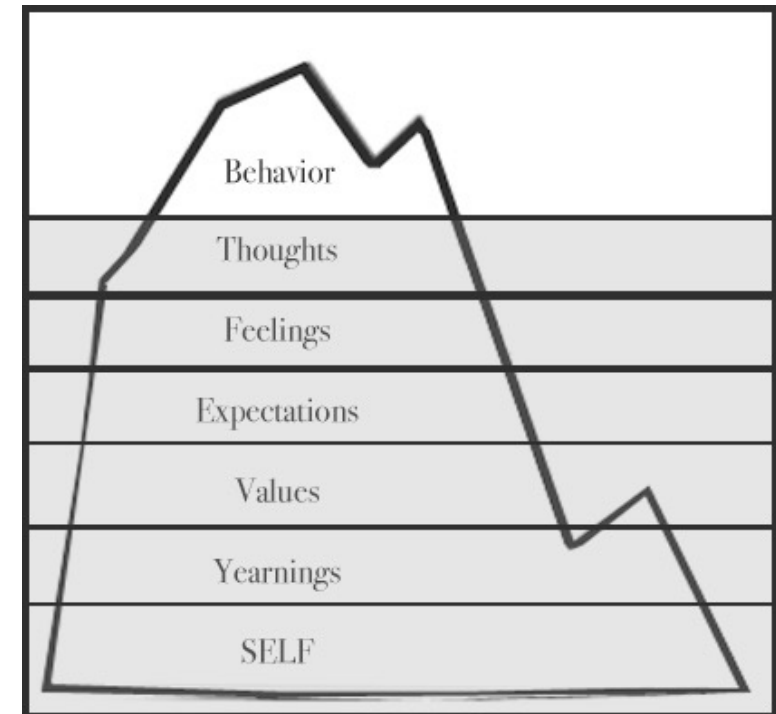
Why do AI models hallucinate?

(1) Pre-training data is noisy: online texts are incomplete or outdated



Common Crawl is old news

Limited information about experiences, norms, perceptions



Why do AI models hallucinate?

(2) Model pre-training is not designed for “factuality”

- Pretraining objectives train models to produce the most plausible text
- The President of the United States of America is <fill the blank>.
 - What is the most likely answer?
- No control in model training for learning or memorizing specific knowledge
- As a result - models generate **highly fluent text** which might or might not be grounded in truth, facts, or knowledge

Why do AI models hallucinate?

Stochastic parrot (Bender et al.)

LLMs generate text by sampling likely word sequences, not by reasoning about truth or meaning

- Outputs are **statistically plausible continuations**, learned from massive text corpora, next word predictions
- Models **do not have grounding** in the external world, intent, or lived experience

“Everything Is a Hallucination”

- Any output is a **probabilistic guess**, not a verified fact
- Even *correct* answers are correct **by chance**
- There is **no distinction** between truth, error, fiction, or speculation

Ethical Implications

- Fluency can **mask uncertainty and error**, increasing over-trust

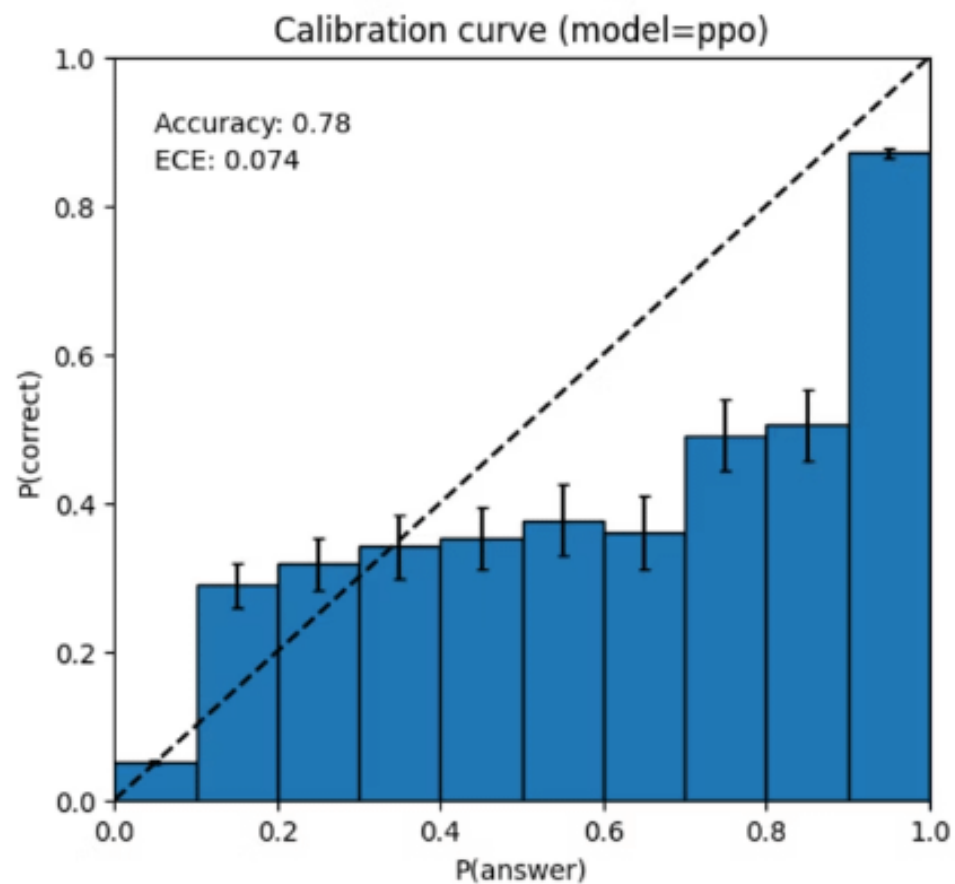
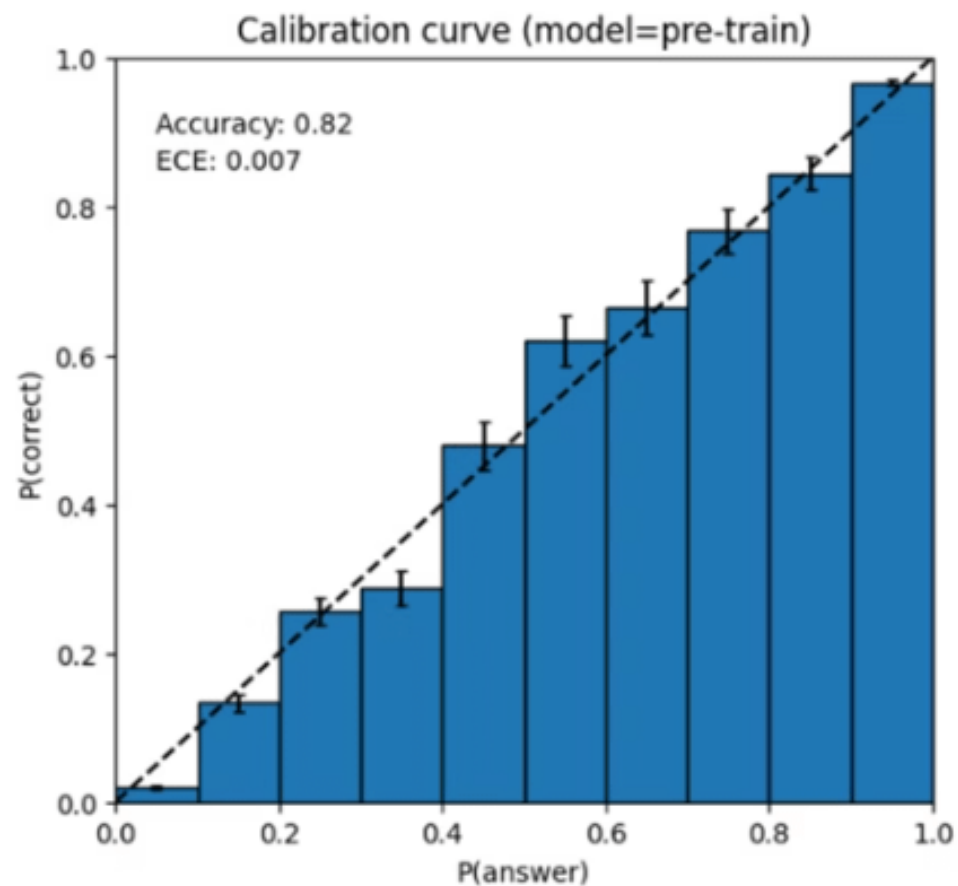


Why do AI models hallucinate?

(3) Post-training alignment via SFT/RLHF can encourage models to diverge from prior knowledge

- AI can be confidently wrong in its predictions. Interestingly, the base pre-trained model is highly calibrated (its predicted confidence in an answer generally matches the probability of being correct).
- However, through current post-training process, the calibration is reduced.

- Calibration plot of the pre-trained GPT-4 model:



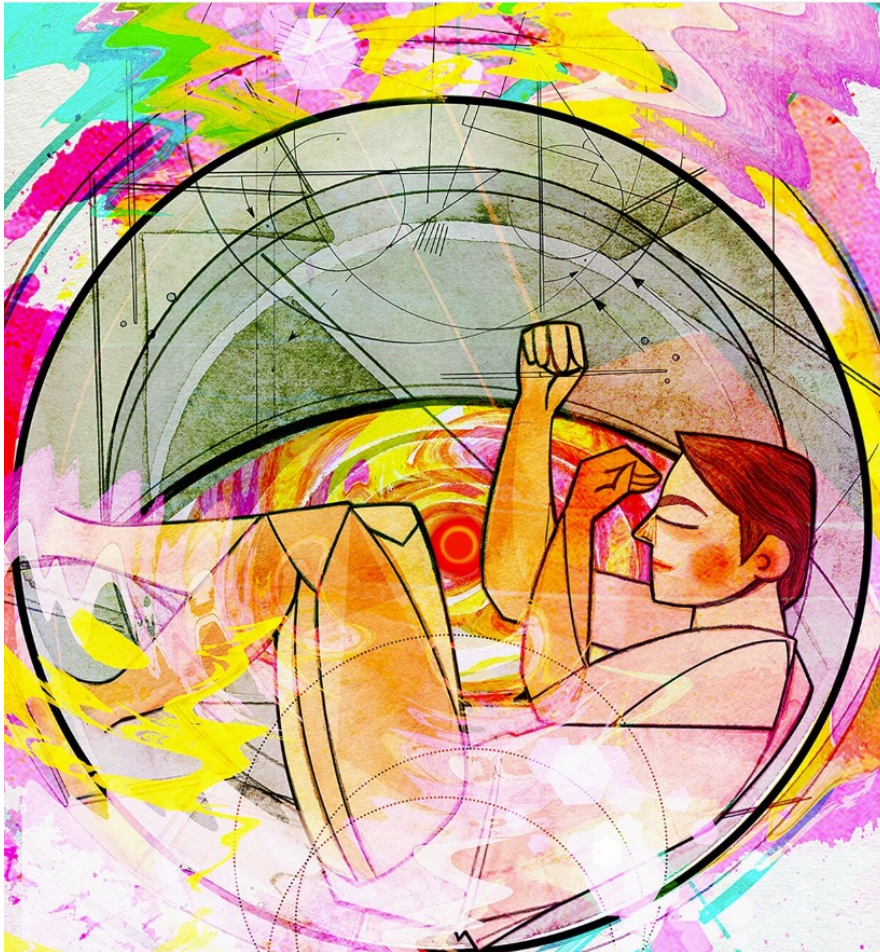
<https://openai.com/index/gpt-4-research/>

Linear Regression also hallucinates!

Every miscalibrated model (and most of models are miscalibrated) that over confidently predicts something with confidence exceeding its actual accuracy is well hallucinating



Is hallucination always bad?



Opinion

Josh Tyrangiel

Honestly, I love when AI hallucinates

December 27, 2023 More than 2 years ago

Intermission:
Should we use the
term “**hallucination**”?



AI anthropomorphism

Dimensions of anthropomorphism that capture uniquely human traits:

- Cognitive states: “AI model **knows**”, “model **understands**”
- Emotional states: “AI model **suffers**”, “model **feels you**”
- Agency: “AI model can book a flight for you”, “ask ChatGPT to **do that for you**”



‘He satisfies a lot of my needs’:
Meet the women in love with
ChatGPT

By Beatrice Nolan
Tech Reporter
December 26, 2023, 3:05 AM ET

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FORTUNE

AI firms warned to calculate threat of
super intelligence or risk it escaping
human control

AI safety campaigner calls for existential threat assessment
akin to Oppenheimer’s calculations before first nuclear test



AI anthropomorphism

Some have criticized the specific term "AI hallucination" for unreasonably anthropomorphizing computers

- Misleading the public
- Dehumanizing humans
- Related ethical concern: Gender bias (female voices for AI assistants), encoding “whiteness”

ai atlas > Tech > Services & Software > AI

We Seriously Need to Stop Anthropomorphizing AI. Here's Why It's Harmful

Commentary: Language shapes public perception. When words are used carelessly or are intentionally anthropomorphic, the public ends up with a distorted picture of what AI can and can't do.



Macy Meyer

Dec. 26, 2025 4:15 a.m. PT

4 min read



Resisting Dehumanization in the Age of ‘AI’

Emily M. Bender

Department of Linguistics

University of Washington

Abstract

The production and promotion of so-called ‘AI’ technology involves dehumanization on many fronts. I explore these processes of dehumanization and the role that cognitive science can play by bringing a richer picture of human cognition to the discourse.

AI anthropomorphism

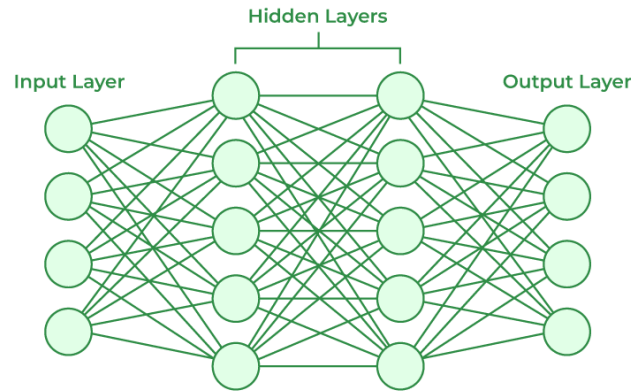
A more serious byproduct of the tendency to talk about machines in anthropomorphic terms is the companion phenomenon of talking about people in mechanistic terminology. The critical reading of articles about computer-assisted learning [...] leaves you no option: in the eyes of their authors, the educational process is simply reduced to a caricature, something like the building up of conditional reflexes. ***Dijkstra, 1985.***

AI anthropomorphism

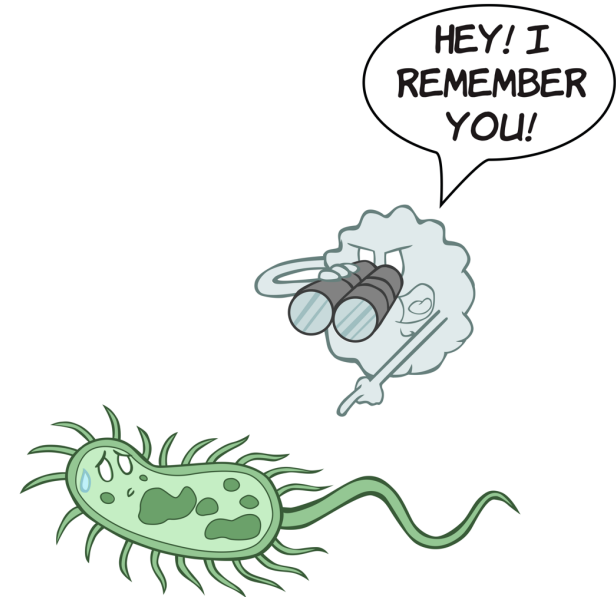
Others argue, metaphorical language is not harmful per se



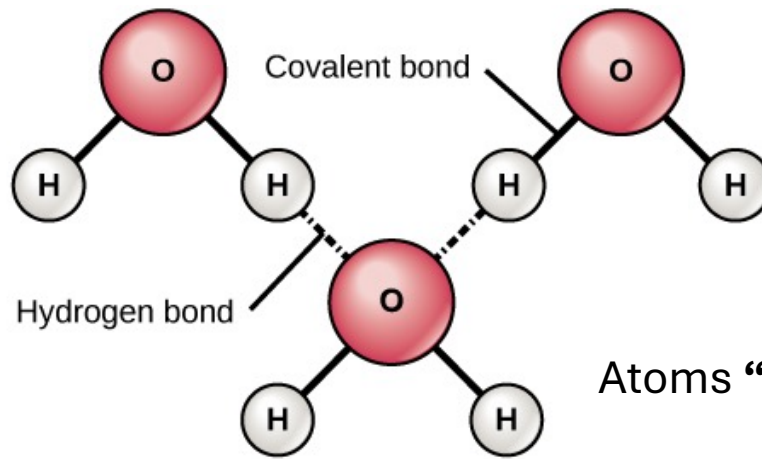
Vaccines **“save lives”**



Model **“learns”**



Immune system **“remembers”**



Atoms **“want to bond”**



In-class activity

Class activity

Should we use anthropomorphizing language (such as “hallucination”) when discussing AI and its ethical implications?

- **10 minutes: Discussion in groups of 4**

1. What does the term “**hallucination**” *help* people understand?
2. What does it *hide or distort*?
3. Who benefits from using anthropomorphic language?
4. Who might be harmed?
5. Would you use this term when:
 - Talking to engineers? Talking to the public? Talking to friends or family? Writing reports?

- Each group should prepare:

- **One strong argument in favor**
- **One strong argument against**

Class activity

Position spectrum



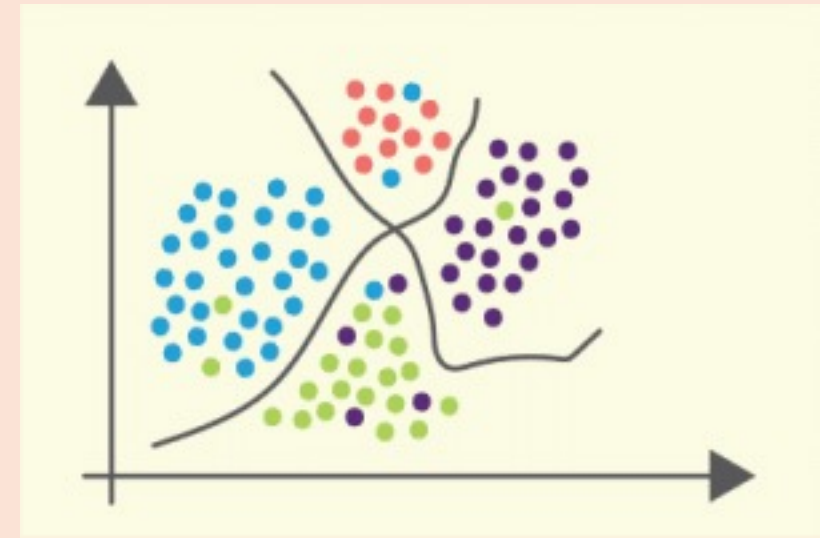
We should never use
anthropomorphizing terms

Anthropomorphizing terms are often
useful and appropriate

End of the intermission:
Back to the topic of
“hallucinations”



How can we detect hallucinations?



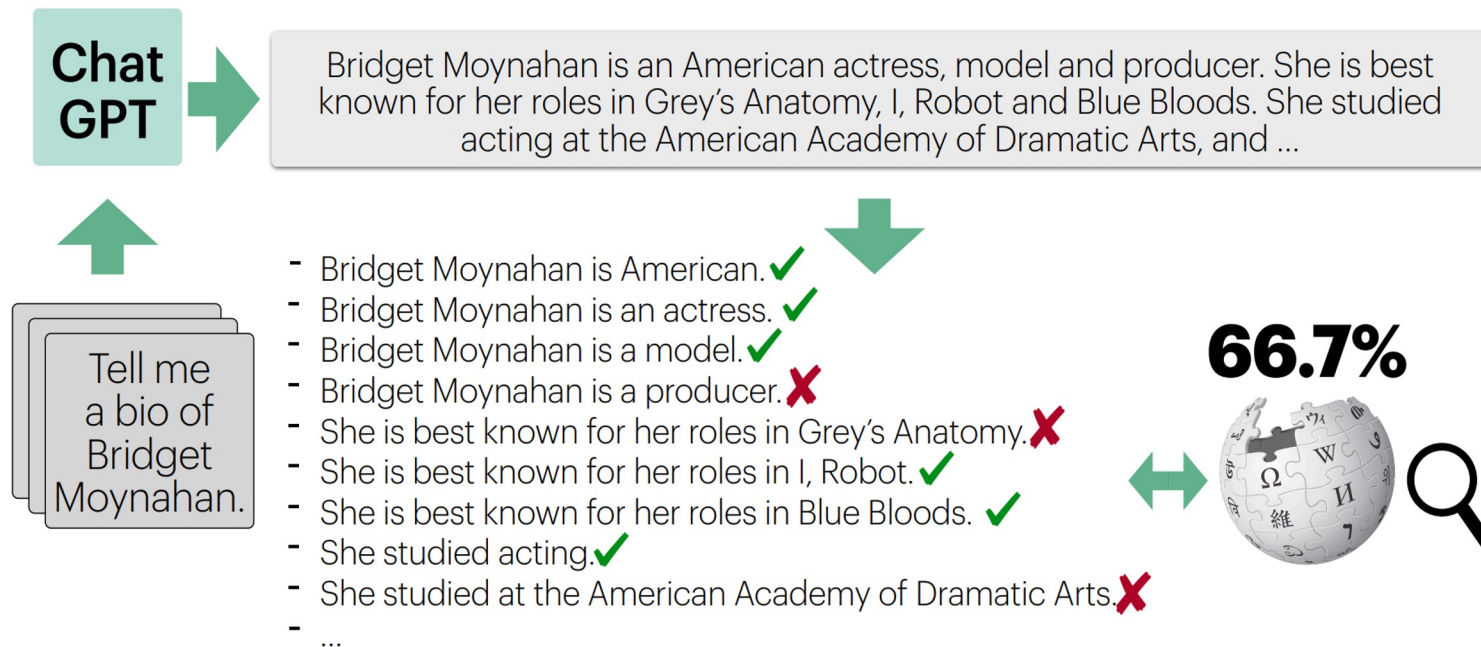
Detecting hallucinations

1. Training custom classifiers for detecting hallucinations
 - Training Data: Generated Text -> Errors Annotated
 - Human Annotated Data
 - Expensive - Long Process to read and edit text
 - Subjective - Factuality decisions have low agreement across annotators
 - Synthetic Data - Create synthetic incorrect text

Detecting hallucinations

2. Retrieve evidence from the web

- Retrieve documents that match the query



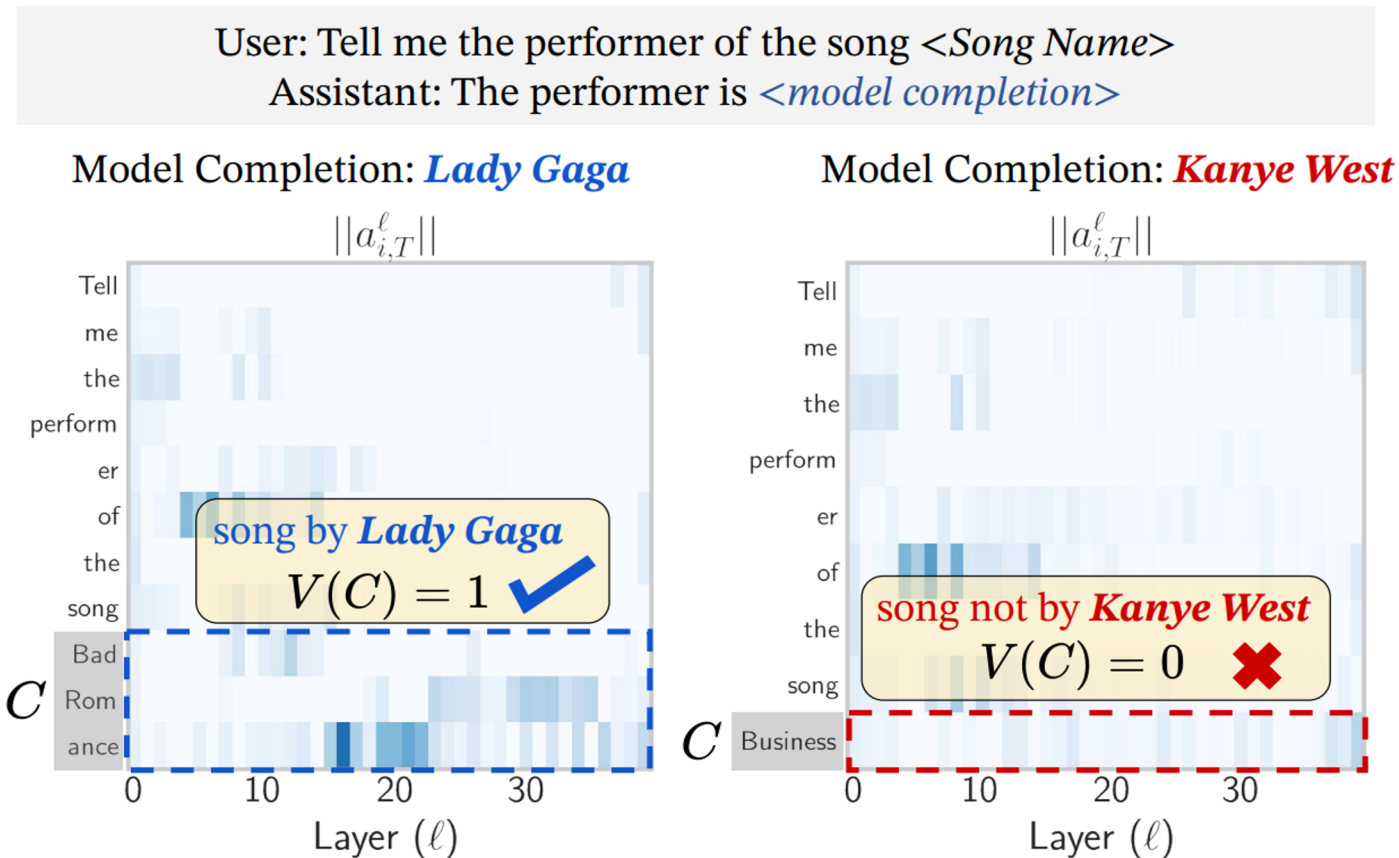
Detecting hallucinations

- **FACTScore** measures the factual accuracy of text generated by LLMs
- Breaks down generated text into atomic facts and calculates the percentage supported by reliable sources
- Provides a fine-grained evaluation compared to binary judgments of quality

Detecting hallucinations

3. Leveraging signal from model internals to detect errors

Model internals like logits, attention, confidence can indicate hallucinations



On Hallucination and Predictive Uncertainty in Conditional Language Generation (Xiao et. al, 2021)

ATTENTION SATISFIES: A CONSTRAINT-SATISFACTION LENS ON FACTUAL ERRORS OF LANGUAGE MODELS (Yuksekgonul et. al, 2024)

Detecting hallucinations

SelfCheckGPT

- A a zero-resource approach designed to detect hallucinations in LLM-generated responses
- The key idea is to use sampling-based methods to evaluate the consistency of generated responses without relying on external databases
- Aims to enhance the reliability and trustworthiness of LLM outputs, especially in scenarios where access to the model's internal states or external databases is not feasible

Detecting hallucinations

SelfCheckGPT

- If an LLM has information about a given concept, sampled responses are likely to be similar and contain consistent fact
- However, for hallucinated facts, stochastically sampled responses (i.e., token sampling methods such as top-p/top-k sampling, adjusting the softmax temperature, etc.) are likely to diverge and contradict one another

How to benchmark hallucination?

Introducing the Hallucination Index



Yash Sheth 
Co-founder and COO



Atindriyo Sanyal 
Co-founder and CTO



Vikram Chatterji 
Co-founder and CEO

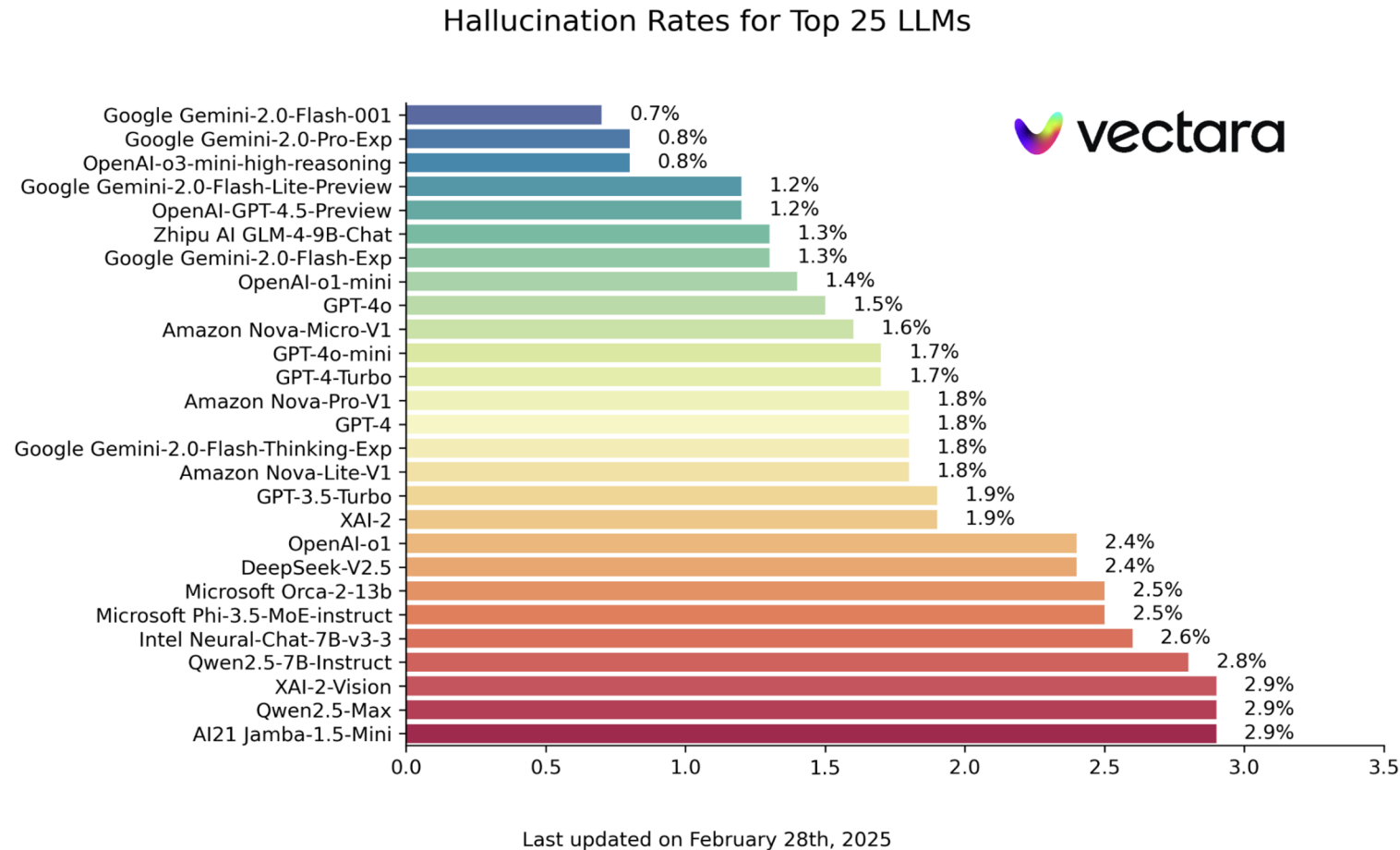
E T I C P R M P T D A T A E T M O O
X P E / M E N T O / N S I H
S I G N / A G L L M E X A L E H M
E L G U E N & P A K I F A L O
C A T N R L U / O T I N I / O X O

LLM Hallucination Index

A Ranking & Evaluation Framework For LLM Hallucinations

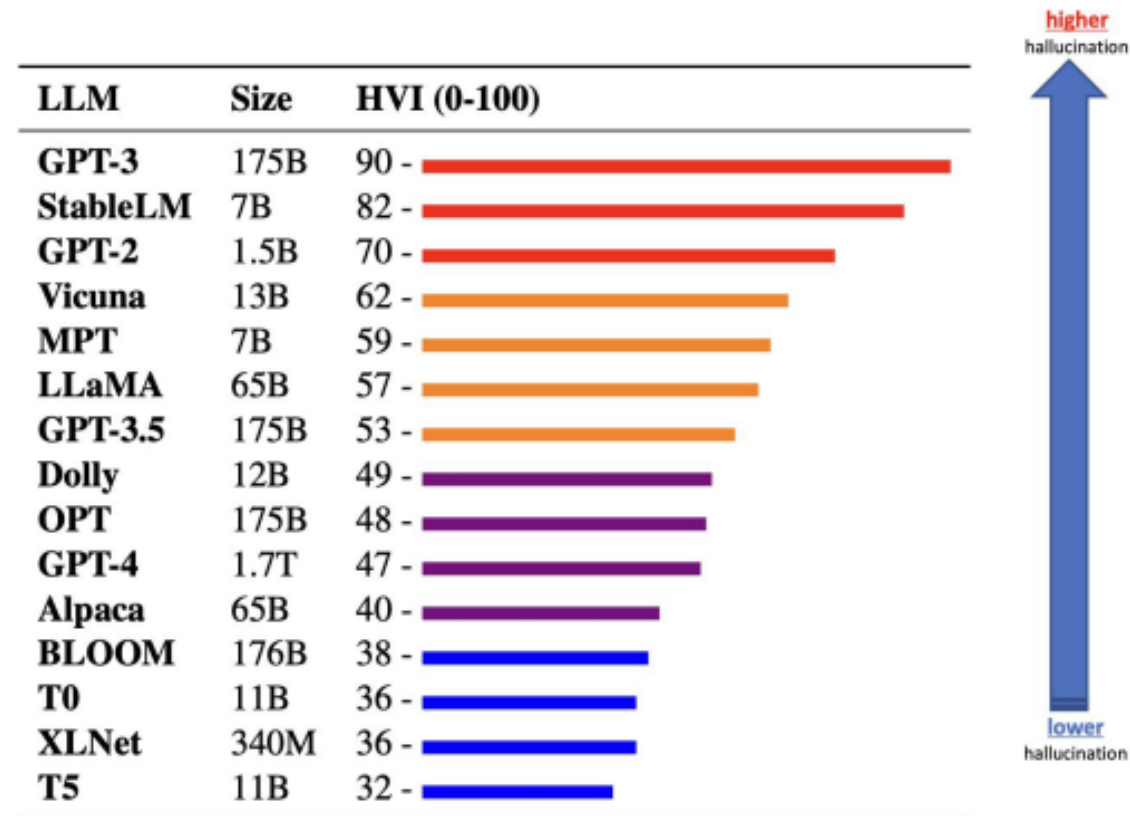
<https://galileo.ai/blog/hallucination-index>

How to benchmark hallucination?



How to benchmark hallucination?

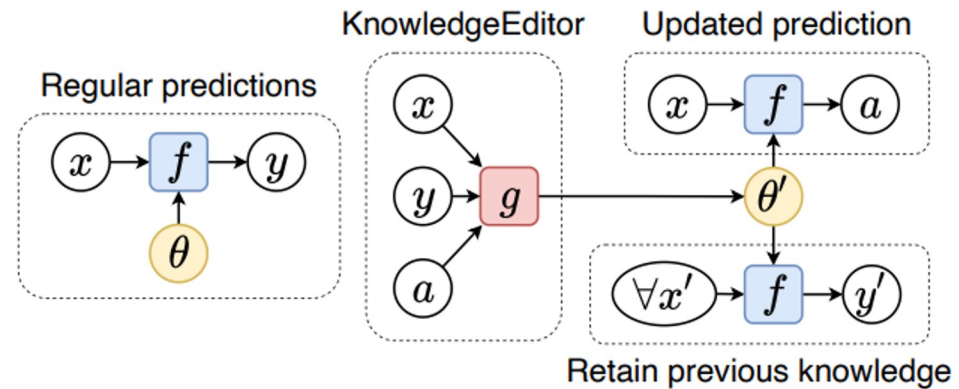
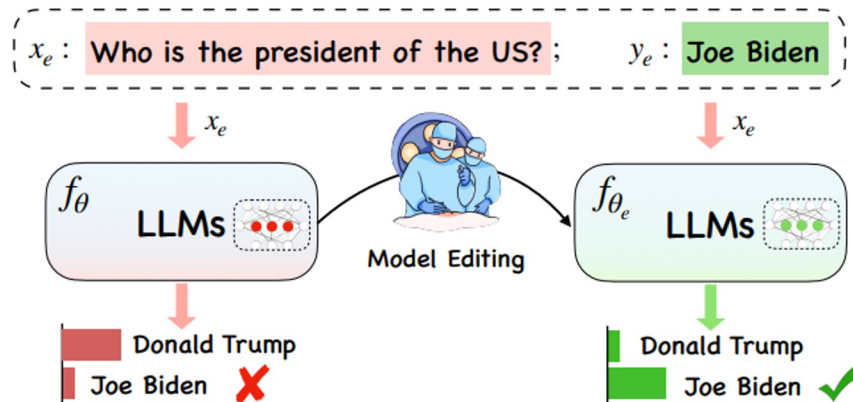
Hallucination Vulnerability Index (HVI)



How can we mitigate?

1. Model editing to model weights for changing facts

- Modifying model weights such that specific facts (outputs) are changed without affecting other facts





2. Back to the root cause

- Pretraining data filtration and clean up can provide largest benefits/impact
- Cleaning the common crawl

3. Post-training fix

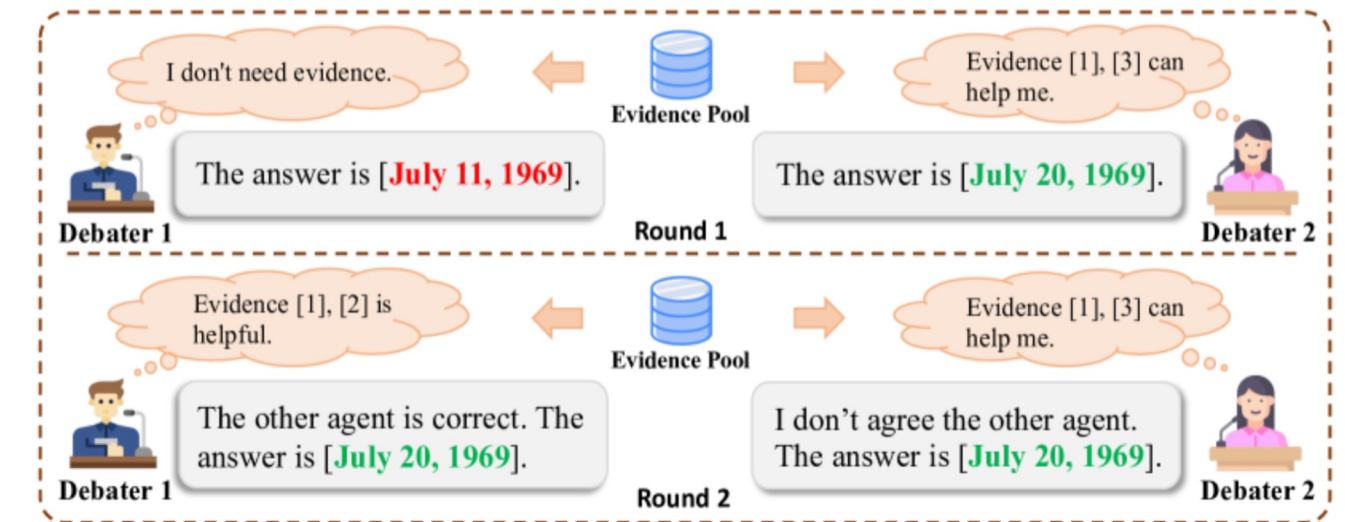
- LLMs without RLHF (Reinforcement Learning from Human Feedback) (Ziegler et al., 2019) tend to exhibit a higher tendency for hallucination
- Align models for factuality based preference data collected from humans or artificially via models/heuristics
- Alignment/RL(?)F/SFT to tune model for improved factuality

4. At inference-time: Self Evaluate, Self-Verify & leverage powerful LLMs for verification

- Ask your LLM itself if it's sure, whether the generation is correct (some prompting expertise might be required)
- Understanding is easier than generation ?

5. Agentic solutions

- Teaching models to abstain when potentially wrong
- Enable LLMs to acknowledge knowledge limitations and abstain from answering
- Augment self-verification with external tools
- Multi-agent debates



Readings for Wednesday

HALoGEN🔑: Fantastic LLM Hallucinations and Where to Find Them

Abhilasha Ravichander^{1*} Shruti Ghela^{1†*} David Wadden² Yejin Choi³

¹University of Washington, ²Google, ³Stanford University

<https://halogen-hallucinations.github.io>

aravicha@cs.washington.edu, yejinc@stanford.edu
{shrutighela1, dave.wadden}@gmail.com

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Patrick Lewis^{†‡}, Ethan Perez⁺,

Aleksandra Piktus[†], Fabio Petroni[†], Vladimir Karpukhin[†], Naman Goyal[†], Heinrich Küttler[†],

Mike Lewis[†], Wen-tau Yih[†], Tim Rocktäschel^{†‡}, Sebastian Riedel^{†‡}, Douwe Kiela[†]

[†]Facebook AI Research; [‡]University College London; ⁺New York University;
plewis@fb.com

RAG (retrieval-augmented generation)

- A hybrid model, combining pre-trained models with non-parametric memory (dense vector index of Wikipedia) to enhance performance on knowledge-intensive tasks
- To improve the accuracy and informativeness of responses generated by NLP models on tasks that require substantial background knowledge, while addressing limitations of current models in accessing and manipulating explicit knowledge

RAG (retrieval-augmented generation)

- RAG is NOT the foolproof solution! Does Fine-Tuning LLMs on New Knowledge Encourage Hallucinations? - <https://arxiv.org/pdf/2405.05904>
- 7 failures of RAG - <https://arxiv.org/pdf/2401.05856>

Post-class activity

Find one real example (news article, blog post, product documentation, or AI tool UI) that uses anthropomorphizing language to describe AI

- Examples: *hallucinating, thinking, lying, understanding, deciding*
- **Quote the specific language** used.
- Answer the following:
 - What does this language make easier to understand?
 - What does it risk obscuring or misrepresenting?
 - Who appears responsible for the AI's behavior *because* of this wording?
 - How might a different wording shift ethical responsibility?
- **Rewrite the passage** using non-anthropomorphic language while preserving clarity.
- **Take a position:** Should anthropomorphizing language be avoided, allowed, or strategically used in this context? Why?

References

- [Koenecke, Allison, et al. "Careless whisper: Speech-to-text hallucination harms." Proceedings of the 2024 ACM conference on fairness, accountability, and transparency. 2024.](#)
- Lee, Nayeon, et al. Factuality enhanced language models for open-ended text generation. Advances in Neural Information Processing Systems 35 (2022): 34586-34599.
- Ladhak, Faisal, et al. When do pre-training biases propagate to downstream tasks? a case study in text summarization. Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics. 2023.
- Vipula Rawte, Swagata Chakraborty, Agnibh Pathak, Anubhav Sarkar, S.M Towhidul Islam Tonmoy, Aman Chadha, Amit Sheth, and Amitava Das. 2023. The Troubling Emergence of Hallucination in Large Language Models An Extensive Definition, Quantification, and Prescriptive Remediations. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pages 2541–2573, Singapore. Association for Computational Linguistics.